PMx Weekly Digest: October 19, 2026

Content Source: Full Text

Paper 1: Integrating a Tumor Size-Overall Survival Model Into a Spatial Quantitative Syst

Bibliographic Information

- **Title**: Integrating a Tumor Size-Overall Survival Model Into a Spatial Quantitative Systems Pharmacology Framework for Antibody-Drug Conjugates in Non-Small Cell Lung Cancer.
- **Authors**: Mehdi Nikfar et al.
- **Journal**: CPT: Pharmacometrics & Systems Pharmacology
- **DOI**: 10.1002/psp4.70352

Paper Type: POPK_PKPD

Quick Take

This paper presents a spatial PBPK-QSP framework that couples whole-body ADC disposition, a Krogh-cylinder tumor model, and a published NSCLC TS-OS model. It is essential reading for QSP/PBPK modelers and ADC translational scientists seeking to link mechanistic tumor dynamics to survival endpoints without patient-level tumor data. The framework reproduces PK, ORR, DCR, and OS trends for T-DXd and Dato-DXd with only two fitted parameters.

Executive Summary

The authors developed an integrated spatial QSP platform that mechanistically links ADC and payload PBPK, intratumoral transport in a Krogh cylinder, tumor cell killing via a transit-compartment PD cascade, and overall survival through a published NSCLC TS-OS model. Using a common model structure and virtual-patient framework, with parameters from literature or independent assays and only two estimated clearance parameters, the model simultaneously reproduced serum PK, ORR, DCR, and exploratory OS for trastuzumab deruxtecan in HER2-mutant NSCLC and datopotamab deruxtecan in EGFR-mutated NSCLC. The work demonstrates transportability of a non-ADC TS-OS model to ADC-generated tumor trajectories and supports model-informed dose selection, identifying 5.4 mg/kg Q3W T-DXd and 6 mg/kg Q3W Dato-DXd as balanced benefit-exposure doses.

Scientific Context & Motivation

ADCs in solid tumors face challenges from heterogeneous target expression, limited tumor penetration, and off-target payload exposure. Existing models separate whole-body PBPK from spatial tumor models, and TS-OS survival models typically require patient-level tumor data. This study addresses the gap by embedding a Krogh-cylinder tumor model within a full-body PBPK framework and linking mechanistically generated tumor trajectories to a published NSCLC TS-OS model developed without ADC-treated populations, enabling early-stage survival projections and dose evaluation.

⚡ Methodological Snapshot

The model integrates a full-body PBPK module for intact ADC, total antibody, and unconjugated payload with a Krogh-cylinder spatial tumor model. The tumor model includes radial diffusion, target binding, internalization, intracellular payload release, bystander killing, and mixed Ag+/Ag– cell populations. A transit-compartment dual PD cascade describes delayed cell killing. Tumor trajectories are linked to OS via a published NSCLC TS-OS model. The system comprises 508 ODEs implemented in SimBiology (MATLAB R2024b). Calibration used particle swarm optimization to estimate two clearance parameters against T-DXd serum PK. Virtual patients were generated by sampling parameters from prespecified distributions with selected correlations.

⌧ Structural Model Breakdown

The model comprises: (1) whole-body PBPK with multiple organs (lung, GI, heart, skin, kidney, spleen, brain, etc.), FcRn recycling, pinocytosis, lymphatic drainage, and lysosomal degradation; (2) explicit deconjugation kinetics linking intact ADC to total antibody and unconjugated payload; (3) a Krogh-cylinder tumor compartment with radial discretization, vascular permeability, diffusion, target binding, internalization, intracellular payload release, and bystander effects; (4) mixed Ag+ and Ag– tumor cell populations with transit-compartment dual PD cascade (proliferative, early quiescent, late quiescent, fully quiescent states); (5) TS-OS sub-model linking simulated tumor size trajectories to survival. Total 508 ODEs.

Detailed Methodological Analysis

Modeling Approach

Spatial PBPK-QSP: whole-body PBPK for ADC/antibody/payload disposition coupled to a Krogh-cylinder tumor transport model and transit-compartment PD cascade; TS-OS linkage using Struemper et al. model. Implemented in SimBiology (MATLAB R2024b).

Data Sources

Published phase I T-DXd PK data (Doi et al.) at 6.4 mg/kg Q3W (n=6) for calibration; clinical benchmarks for ORR/DCR/OS from DESTINY-Lung01 (T-DXd) and TROPION-PanTumor01 (Dato-DXd). All other parameters from literature or independent assays.

Estimation Methods

Particle swarm optimization (PSO) in MATLAB/SimBiology fitting framework; two parameters estimated: nonspecific antibody uptake (CL) and unconjugated payload hepatic clearance (CL_PL).

Model Evaluation

Visual comparison of simulated vs observed serum PK; comparison of predicted ORR/DCR and OS curves against clinical benchmarks; VP ensemble simulations (1000-5000 VPs) with median and 95% CI; sensitivity analyses of VP parameters vs response and OS.

Covariate Analysis

VP parameter distributions compared between responders and non-responders; associations between tumor growth rate, HSCR, AgPS and response/OS examined. No formal covariate model; VP parameters sampled from correlated distributions.

Statistical Rigor Assessment

The calibration is based on a small PK dataset (n=6) and only two parameters are estimated, so full model identifiability is not claimed. VP simulations use 1000-5000 virtual patients with repeated sampling to quantify cohort-level uncertainty. ORR/DCR are compared as population-level benchmarks, not fitted. The TS-OS model is applied with an adjusted dispersion parameter. Sensitivity analyses are descriptive rather than formal global sensitivity analyses. Missing individual-level tumor and survival data limit validation.

☒ Key Findings

The calibrated model captured T-DXd serum PK (intact ADC, total antibody, unconjugated payload) with only two fitted parameters: nonspecific antibody uptake (CL) and unconjugated payload hepatic clearance (CL_PL). Simulated ORR and DCR matched clinical benchmarks for both ADCs. Predicted median OS was consistent with DESTINY-Lung01 and TROPION-PanTumor01. Tumor growth rate was the strongest survival-associated input; HSCR and AgPS were associated with response but with overlapping responder/non-responder distributions. Dose-response analysis showed diminishing efficacy gains at higher doses: T-DXd ORR increased from 31% at 4.4 mg/kg to 48% at 5.4 mg/kg and 56% at 6.4 mg/kg, with median OS 14.0, 16.9, and 17.8 months, while payload AUC increased ~23% and ~45%. Dato-DXd ORR increased from 24% at 4 mg/kg to 41% at 6 mg/kg and 51% at 8 mg/kg, with median OS 12.0, 16.3, and 17.0 months, while payload AUC increased ~50% and ~100%. The model supports 5.4 mg/kg Q3W T-DXd and 6 mg/kg Q3W Dato-DXd.

💡 Clinical & Regulatory Implications

The framework supports model-informed dose selection by balancing efficacy (ORR, OS) against unconjugated payload exposure as a safety surrogate. It identifies 5.4 mg/kg Q3W T-DXd and 6 mg/kg Q3W Dato-DXd as balanced doses, consistent with approved doses. It also highlights the need for multidimensional biomarkers combining tumor growth rate, HSCR, and AgPS to predict response. The platform can be used to explore ADC design levers (DAR, linker, payload) and to generate hypotheses for future clinical validation.

Strengths & Limitations

Strengths

- First framework linking whole-body ADC PBPK, spatial Krogh-cylinder tumor model, and OS in a single structure

- Only two parameters estimated from serum PK; all other parameters from literature or independent assays
- Common model structure and virtual-patient framework successfully applied to two ADCs with distinct targets
- Reproduces PK, ORR, DCR, and OS trends without fitting individual tumor trajectories or adding ad hoc response compartments
- Provides a practical demonstration of TS-OS model transportability to ADC-generated tumor trajectories
- Model-informed dose selection framework jointly evaluates efficacy and exposure-based safety surrogates

Limitations (Acknowledged by Authors)

- Tumor burden represented as a single lumped lesion without lesion-level heterogeneity or new lesions
- ORR/DCR evaluated at fixed 12-month landmark rather than RECIST-based BOR
- Toxicity not modeled mechanistically; payload AUC used as exposure-based surrogate
- Predicted human tissue profiles not validated against clinical tissue data
- TMDD not included; target expression in healthy tissues omitted
- Obesity-related physiological differences not represented (proportional scaling of all tissues with body weight)
- TS-OS application required adjustment of survival dispersion parameter
- OS predictions are empirical translational projections, not mechanistic survival model

Limitations (Expert Review)

- Only two parameters estimated; identifiability of the full 508-ODE system is not established
- Use of T-DXd PK data from non-NSCLC patients (breast/gastric) to inform NSCLC exposure may introduce bias
- Common DXd payload clearance assumed for both ADCs, potentially affecting payload exposure predictions
- Responder/non-responder overlap indicates limited standalone predictive value of individual biomarkers
- Fixed spatial parameters (capillary radius, diffusivity) across virtual patients may underestimate inter-patient variability in tumor physiology
- Simulations of 5000 VPs with repeated sampling may not fully capture parameter uncertainty propagation

Generalizability

The framework is designed to be transferable across ADC formats and targets, as demonstrated by application to T-DXd and Dato-DXd. However, generalizability to other tumor types, payloads, or combination regimens requires re-parameterization and validation. The TS-OS linkage is specific to NSCLC and may not extrapolate to other indications.

Figures & Tables

- **Figure 1**: ADC Spatial PBPK-based QSP model: (a) anatomical diagram, (b) tissue PK, (c) tumor PK, (d) PD model.
  - *Significance*: Provides the overall model structure and illustrates the integration of whole-body PBPK, Krogh-cylinder tumor, and PD cascade.
- **Figure 2**: PK calibration: (a) observed and simulated serum concentration-time profiles of intact ADC, total antibody, and unconjugated payload; (b) optimization distributions; (c) predicted tissue payload AUC.
  - *Significance*: Demonstrates the calibration of the model to clinical PK data and the predicted tissue exposure profiles.
- **Figure 3**: Model-predicted tumor response trajectories and efficacy endpoints for T-DXd and Dato-DXd: (a,b) simulated percent change SLD; (c,d) waterfall plots; (e,f) ORR/DCR vs clinical benchmarks.
  - *Significance*: Shows the model's ability to reproduce ORR and DCR for two ADCs.
- **Figure 4**: OS predictions for T-DXd and Dato-DXd: (a,b) OS curves from 5000 VPs; (c,d) comparison with clinical survival curves.
  - *Significance*: Demonstrates the TS-OS linkage and transportability of the survival model.
- **Figure 5**: Comparison of VP parameter distributions between responders and non-responders for T-DXd.
  - *Significance*: Illustrates limited standalone discrimination of biomarkers.
- **Figure 6**: Association between VP parameters and median OS, colored by maximum tumor size reduction.
  - *Significance*: Shows how depth of response modulates survival for given parameter values.
- **Figure 7**: Model-informed dose-response and exposure trade-offs for T-DXd and Dato-DXd: (a) ORR, (b) median OS, (c,d) tissue and plasma payload AUC.
  - *Significance*: Supports dose selection by balancing efficacy and payload exposure.
- **Table S1**: Parameter values and sources for the PBPK-QSP model.
  - *Significance*: Provides the full parameterization and sources, enabling reproducibility.

Code & Reproducibility Assessment

The complete MATLAB SimBiology platform was developed within GlaxoSmithKline and cannot be publicly shared under company policy. Supporting Information provides full model equations, structure, assumptions, parameter definitions and sources, and calibration procedures. Additional nonconfidential materials may be considered upon reasonable request subject to GSK review.

Supplementary Materials

Supporting Information includes Data S1 (model equations, compartment definitions, assumptions, parameter sources, calibration details) and Data S2 (Krogh-cylinder geometry, radial transport equations, VP parameter distributions and correlations), plus Table S1 with parameter values and sources.

Future Directions

Future work should evaluate the framework using matched individual-level PK, biomarker, longitudinal tumor-size, treatment, safety, and survival data. Incorporating prior treatment, performance status, inflammatory markers, and tumor-burden biomarkers may improve patient heterogeneity representation. Extensions could include explicit toxicity modules (QST), immune components, lesion-level dynamics, and multidimensional biomarker identification.

Expert Commentary

This paper represents a significant step toward integrating mechanistic spatial tumor models with clinically meaningful survival endpoints. The use of a published non-ADC TS-OS model to translate QSP-generated tumor trajectories is pragmatic and addresses a common translational bottleneck. However, the reliance on aggregate clinical benchmarks and the small number of fitted parameters means the model should be viewed as a hypothesis-generating platform rather than a validated predictive tool. The dose-selection framework is particularly useful, as it explicitly quantifies the trade-off between efficacy gains and payload exposure. The next frontier is prospective validation with individual-level data and incorporation of toxicity and immune modules.

Bottom Line

This study provides a comprehensive spatial PBPK-QSP platform that bridges ADC systemic disposition, intratumoral transport, tumor response, and survival. For practicing pharmacometricians, the key takeaway is that a mechanistically grounded, spatially resolved tumor model can generate clinically consistent tumor trajectories and OS projections without patient-level tumor data, and can support dose selection by balancing efficacy against payload exposure. The framework is a valuable template for early ADC development, though prospective validation with matched individual-level data is needed.

Figures

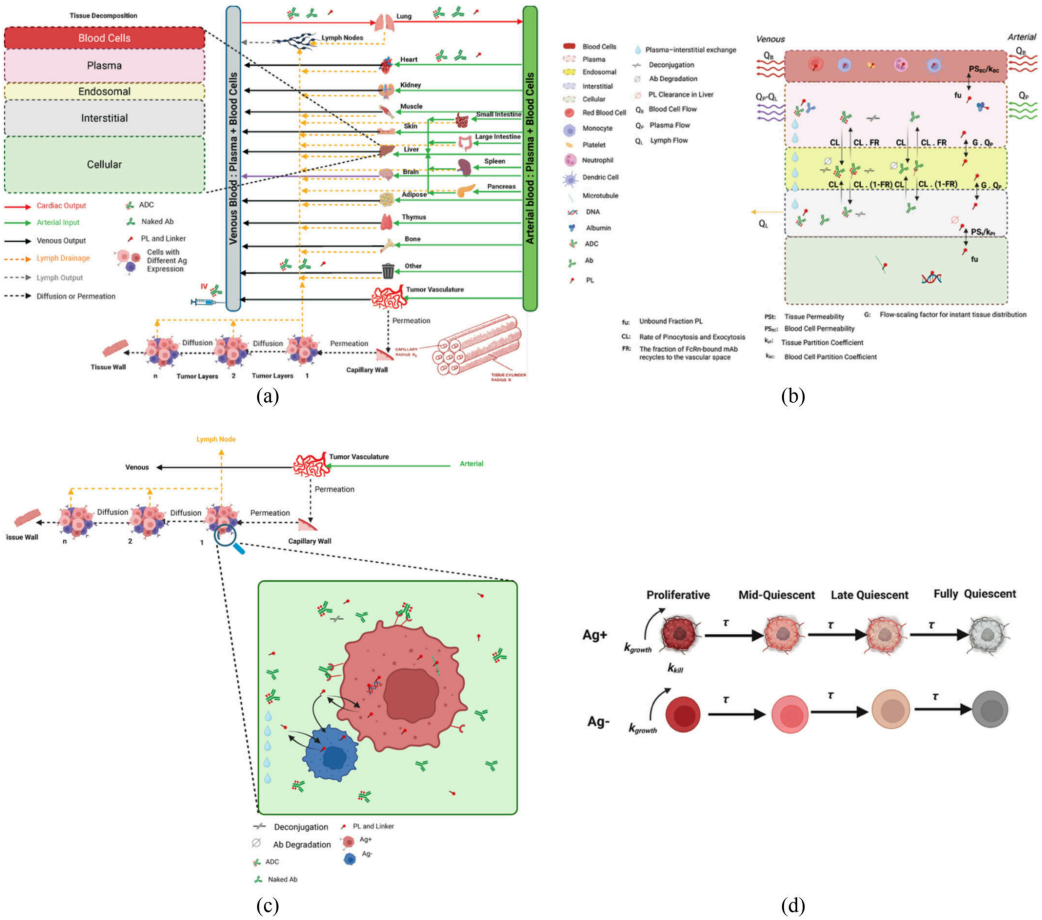


Figure 1: ADC Spatial PBPK-based QSP model, (a) Anatomical diagram, (b) Tissue PK, (c) Tumor PK, (d) PD Model.

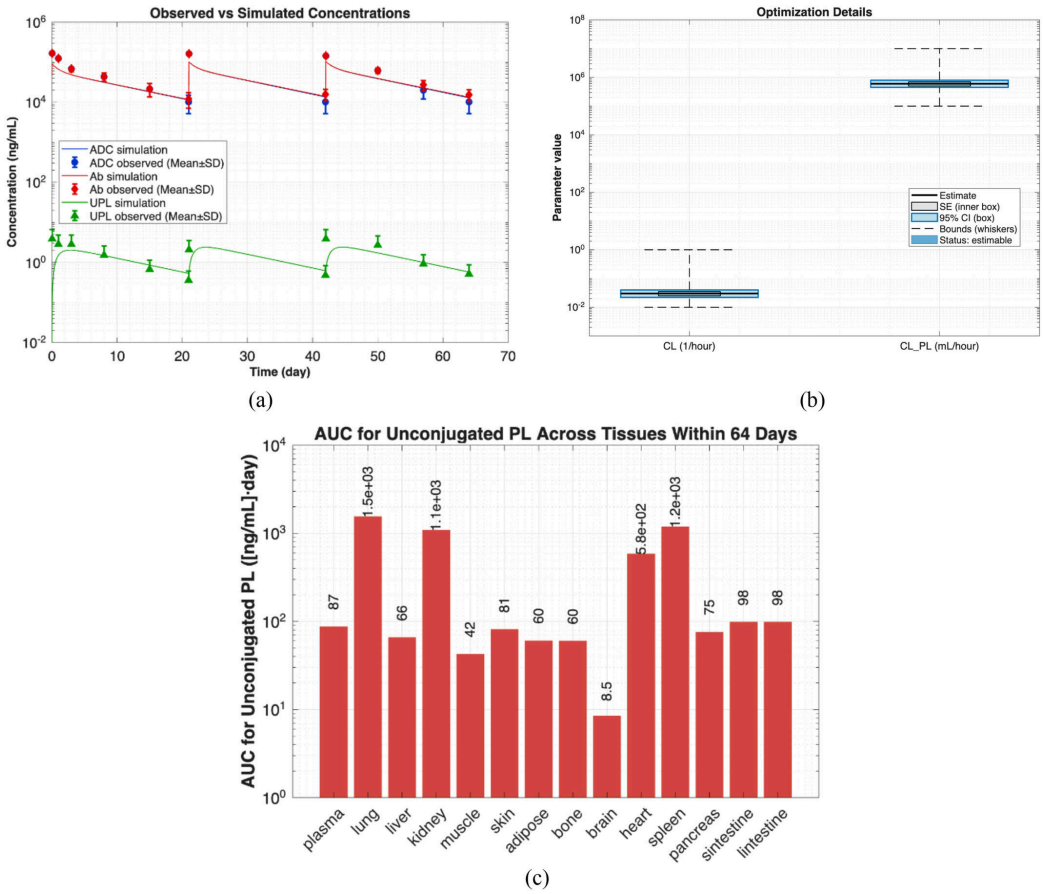


Figure 2: PK calibration. (a) Observed and simulated serum concentration-time profiles of intact ADC, total antibody, and unconjugated payload. (b) Optimization distributi

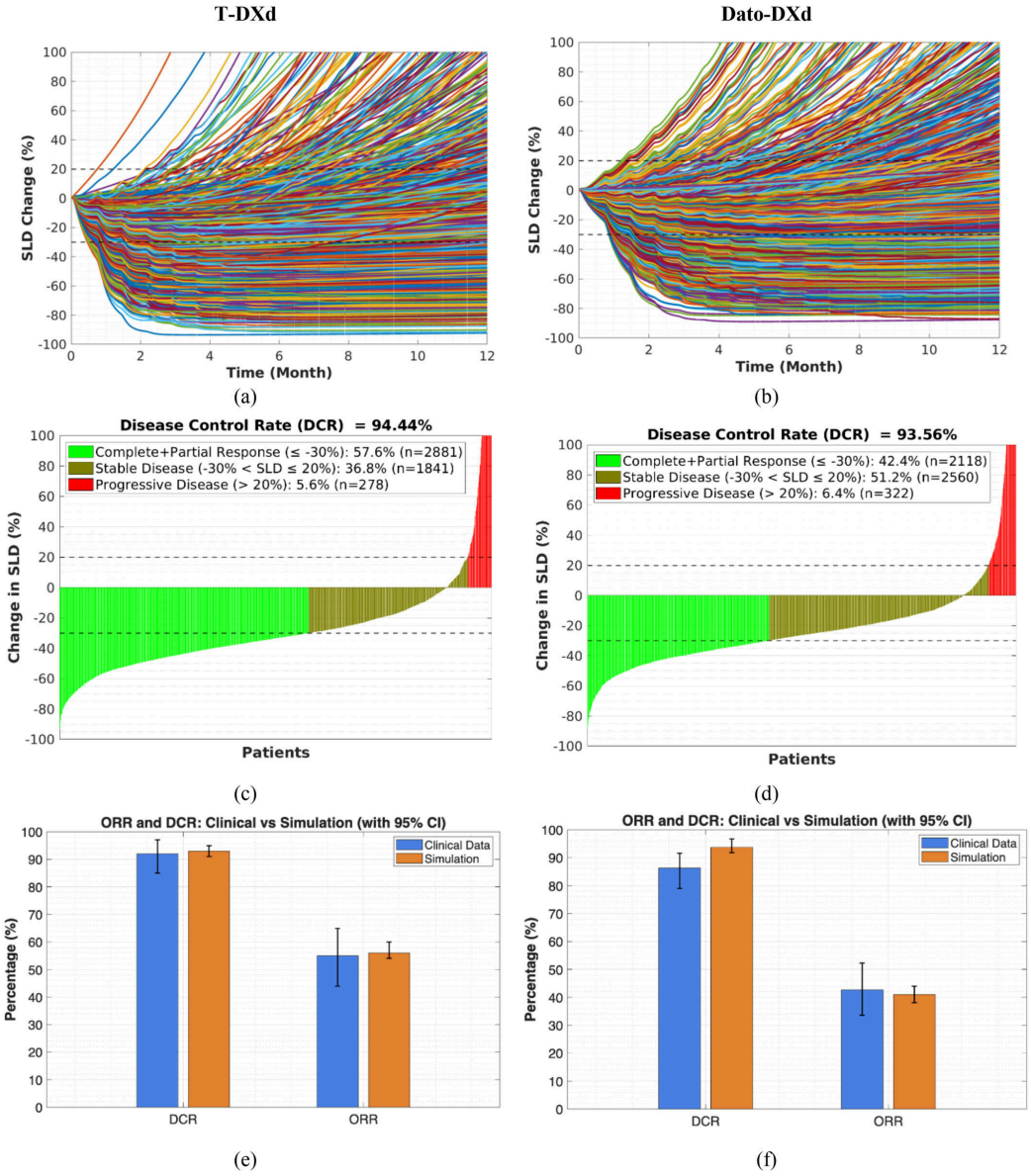


Figure 3: Model-predicted tumor response trajectories and efficacy endpoints for T-DXd (6.4 mg/kg Q3W, left column) and Dato-DXd (6 mg/kg Q3W, right column): (a, b) Simula

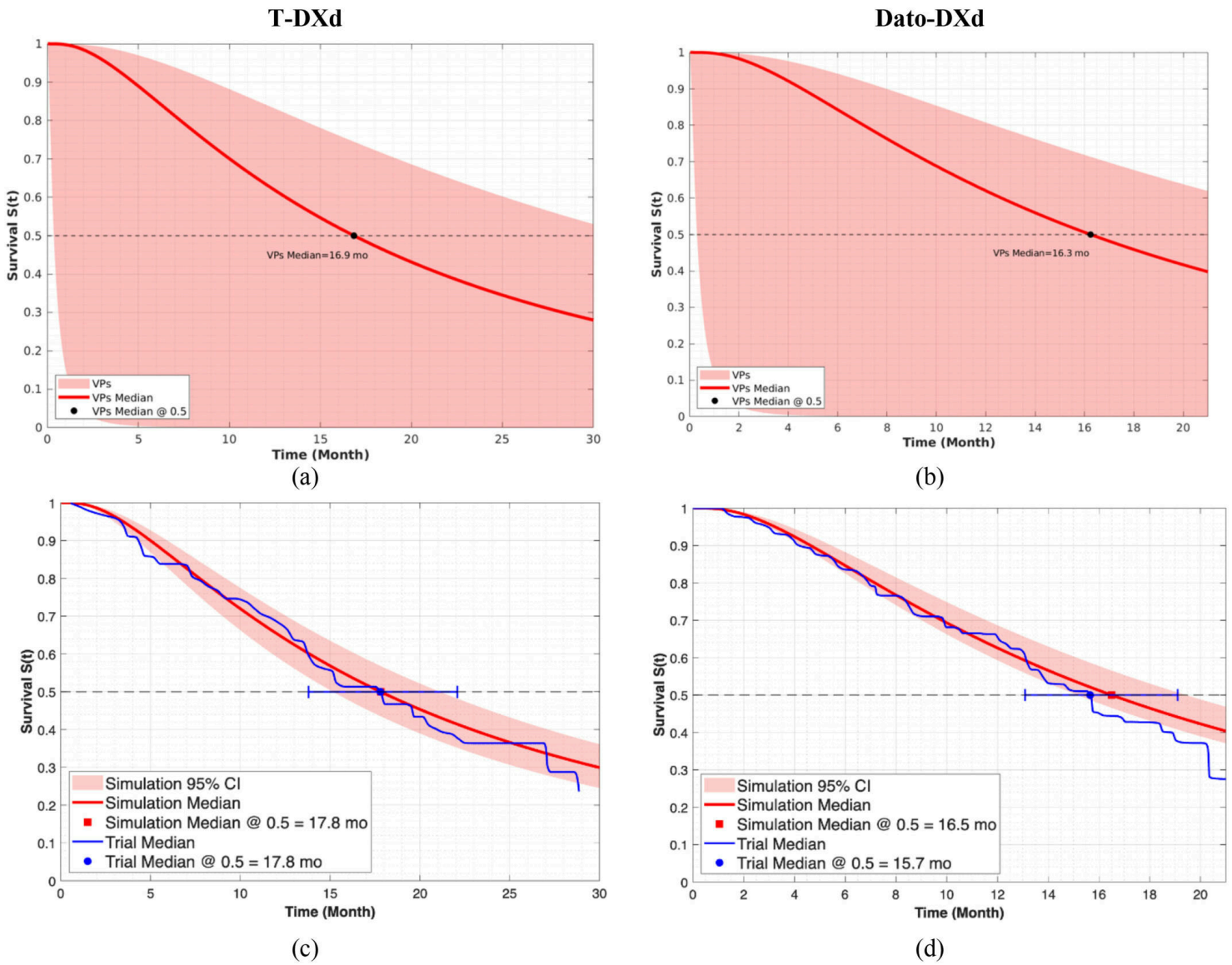


Figure 4: OS predictions for T-DXd (left column) and Dato-DXd (right column): (a, b) OS curves derived from tumor response trajectories simulated in 5000 VPs treated with

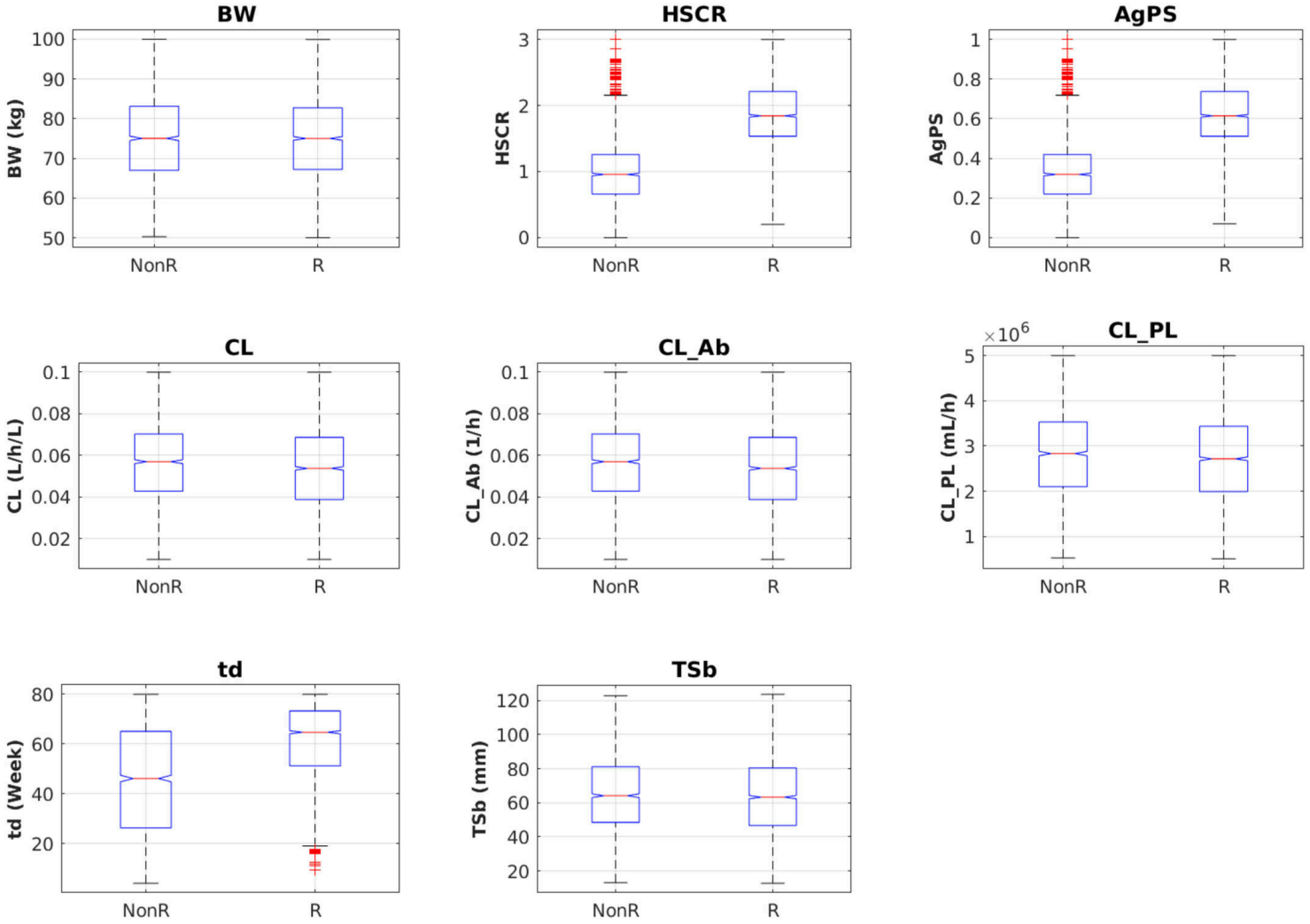


Figure 5: Comparison of VP parameter distributions between responders (R) and non-responders (NonR) in 5000 VPs for T-DXd at 6.4 mg/kg Q3W.

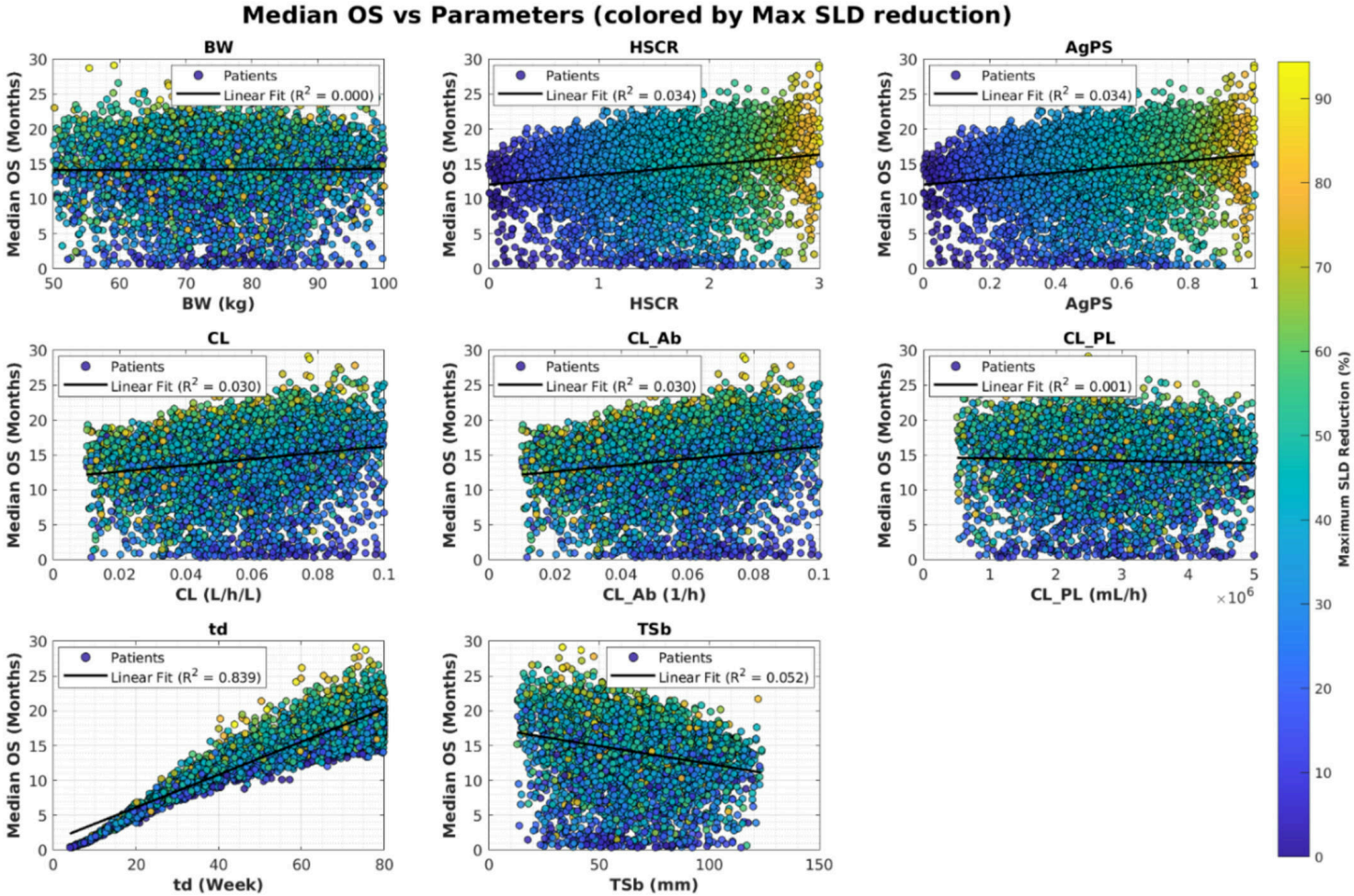


Figure 6: Association between virtual patient parameters and median overall survival, with points colored by maximum tumor size reduction in 5000 VPs for T-DXd at 6.4 mg/k

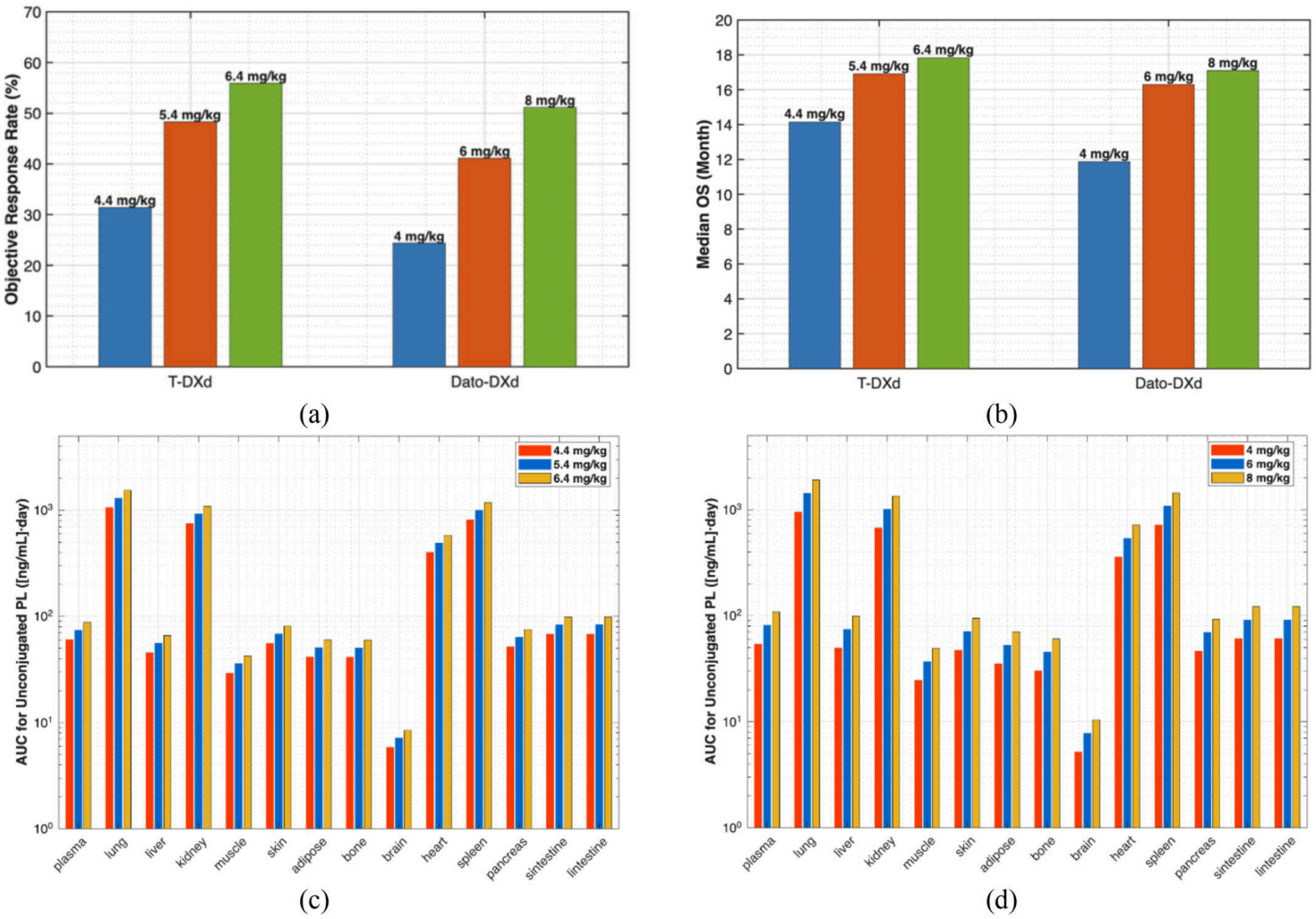


Figure 7: Model-informed dose–response and exposure trade-offs for T-DXd and Dato-DXd in NSCLC: (a) ORR, (b) Median OS, (c) T-Dxd tissue and plasma AUC of unconjugated pay